Heritage Guard

A Living Heritage Intelligence and Safeguarding Atlas for India

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The national-level thesis

Most heritage platforms help people discover a place. Heritage Guard helps a place remain legible, lived, and protectable.

It turns a cultural asset into a source-labelled digital twin: history and living practice for showcasing, geometry and condition for understanding, and evidence workflows for responsible safeguarding.

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1 Executive decision

Heritage Guard should not be pitched as a tourism app with a reporting button. That framing places the team in the largest and least defensible comparison set. The stronger thesis is a **living heritage intelligence layer**: a product that makes one Indian heritage cluster understandable in its historical, geographic, community, and present-condition context, then converts trustworthy public observations into review-ready safeguarding evidence.

The product is intentionally narrow at the first release. It begins with one selected heritage cluster and one validated data pack. The cluster is not merely a place on a map; it is a connected set of tangible assets, local names, cultural practices, oral context, access paths, sensitive zones, and evidence of change. This is the product's central design decision. A generic app accumulates pages. Heritage Guard builds a defensible, versioned record of **what the asset is, why it matters, what is changing, and how certain we are**.

One-line pitch: Heritage Guard is a provenance-aware digital twin for Indian heritage clusters that showcases living cultural context and gives communities a safe, uncertainty-aware way to document changes before they become irreversible.

Decision	Why it wins at SIH	What the prototype proves
Keep the protection niche	It avoids competing with tourism, AR, museum, and quiz apps.	A live geospatial evidence journey.
Make showcasing structural	The asset, local knowledge, and living practice are the first screen, not decoration.	A Heritage Twin with a cultural-context layer.
Add a change ledger	Preservation becomes measurable rather than a one-time complaint.	Before/after observations, condition events, and review status.
Show uncertainty openly	Trust is a product feature and a technical differentiator.	GPS accuracy, boundary version, source confidence, and reasoned outcomes.
Scale through a data contract	The team can present a national roadmap without pretending to have a national database.	One-site MVP with a repeatable onboarding protocol.

2 The opportunity: from "heritage content" to "heritage continuity"

India's heritage is both tangible and living. A monument, shrine, craft street, festival route, stepwell, historic market, or landscape carries meaning through architecture, memory, ritual, language, and everyday use. Official mapping infrastructure already demonstrates the value of locating heritage assets; for example, the Bhuvan/NRSC interface presents an Archaeological Survey of India heritage-sites map with multiple map layers and an explicit data limitation notice [2]. The gap is not another map pin. The gap is a transparent bridge from **knowledge** to **continuity**.

The current reporting pattern is fragmented: a resident sees a change, a visitor takes a photograph, a researcher records a note, or a culture bearer explains a practice. These fragments rarely share a common site identity, coordinate quality, source trail, consent state, or review status. Heritage Guard makes those fragments interoperable without turning community members into investigators or replacing the competent authority.

The official SIH framing asks for ideas that showcase the rich cultural heritage and traditions of India. The response therefore must showcase more than a monument photograph. Heritage Guard showcases the **relationship** between place and practice, then demonstrates why that relationship deserves careful protection. This is consistent with UNESCO's emphasis on the participation of the communities, groups, and individuals who create, maintain, and transmit intangible cultural heritage [4] [5].

2.1 The sharply defined problem

A heritage asset can be visible yet poorly understood, mapped yet poorly contextualized, celebrated yet progressively altered. The practical problem is:

> **When a cultural asset or its living context changes, the public lacks a trusted way to connect cultural meaning, exact place, time, evidence, source provenance, and uncertainty into one reviewable record.**

This is not an accusation problem. It is an **evidence continuity problem**. The product must preserve the difference between a community observation, a map indication, a curator's review, and an authority's finding.

2.2 Why existing categories do not solve it

Category	What it optimizes	What it usually lacks
Tourism guide	Discovery, itineraries, ratings	Protected geometry, evidence provenance, safeguarding workflow.
Virtual museum	Narrative and visual immersion	Current-condition observations and accountable review.
Complaint portal	Submission volume	Heritage context, boundary logic, cultural consent, uncertainty.
Generic AI app	Fast answers or labels	Explainable evidence chain and authority boundaries.
Static heritage map	Location and lookup	Living practice, change history, community observation, case status.

3 Product thesis: the Heritage Twin

The revised product is organized around a reusable object called the **Heritage Twin**. It is a compact, versioned representation of a heritage cluster with five connected layers. This makes showcasing and safeguarding the same product rather than two unrelated modules.

Layer	What it contains	Why it matters
1	Cultural identity	Official name, local name, language, origin story, significance, photographs, and access context.

Layer	What it contains	Why it matters
2	Living practice	Community-authored or consented descriptions of rituals, craft, food, performance, route, or memory associated with the place.
3	Spatial truth	Source-labelled asset geometry, relevant zones, access paths, sensitive areas, and known limitations.
4	Condition and change	Time-stamped observations, structured concern categories, before/after evidence, and a non-accusatory change ledger.
5	Safeguarding workflow	Case ID, provenance, uncertainty, reviewer notes, referrals, and a visible status timeline.

The first two layers satisfy the showcase requirement in a culturally meaningful way. The last three make the product technically distinctive and operationally useful. A visitor can explore; a student can document; a culture bearer can add context; a curator can review; and a future authority can decide.

3.1 The memorable "before it disappears" moment

The demo should show one heritage cluster in two views. The first is the **living view**: the site, its story, local vocabulary, associated practice, and the people or routes that give it meaning. The second is the **continuity view**: a map with the source-labelled boundary, a new observation, GPS accuracy, a change ledger, and a review packet. The emotional insight is simple: preservation begins when the public can see not only what a place looks like, but what would be lost if its context changed.

4 What we build: a focused MVP with national architecture

4.1 MVP promise

The MVP will use one locally accessible heritage cluster, one source-checked spatial layer, one consented cultural-context pack, and four repeatable test scenarios. It must prove the complete chain from showcase to safeguarding in a controlled demo. The team should not claim India-wide coverage, live government integration, or legal verification.

Module	Prototype behavior	Judge-visible proof
Heritage Twin	A rich site page combining history, local identity, cultural practice, map, source, and consent labels.	Showcase is first-class.
Field capture	Photo, neutral observation, category, language, timestamp, location, and device-reported accuracy.	Real-world usability.

Module	Prototype behavior	Judge-visible proof
Spatial reasoning	Point-in-polygon, distance-to-boundary, accuracy-circle overlap, and source-confidence checks.	Nontrivial engineering.
Change ledger	Every observation becomes a dated event linked to a site, layer version, and evidence.	Continuity, not a complaint inbox.
Evidence packet	A reviewable HTML/PDF-style packet with map, source, image, uncertainty, and audit trail.	Institutional utility.
Review console	Case status, reviewer note, request for more evidence, referral, and closure reason.	Accountability without overclaiming.

4.2 What the user sees

The public journey is deliberately simple. A person opens the selected Heritage Twin, learns why the place matters, explores the map and living context, and selects "Document a change." The form asks for what was observed rather than who is guilty. Location capture preserves the raw accuracy value. The result screen separates three statements: "the user reported," "the map indicates," and "an authorized reviewer must verify." The case then becomes a private-by-default change event with a traceable ID.

The reviewer journey is equally important. A curator sees the cultural asset, evidence image, map layer version, source link, accuracy circle, uncertainty reason, and prior events in one packet. They can request additional information, mark a report as reviewed, or refer it through an authorized process. No status label implies a legal conclusion.

5 Technical innovation that is easy to demonstrate

5.1 1. Provenance-aware spatial reasoning

A centre-point radius is not enough for irregular heritage sites. Heritage Guard uses GeoJSON geometry, point-in-polygon testing, distance to the relevant boundary, and a configurable uncertainty rule. If the device accuracy circle intersects the boundary, or if the source is weak or stale, the system refuses a definitive spatial classification. A simplified prototype rule is:

```
if source_confidence < minimum:
    result = "Source insufficient for classification"
else if gps_accuracy_m > configured_limit:
    result = "Location uncertain"
else if distance_to_boundary_m <= gps_accuracy_m + safety_margin_m:
    result = "Boundary uncertain"
else if point_in_polygon(report_point, selected_zone):
    result = "Potential zone concern — authority verification required"
else:
    result = "Outside selected source layer — other concerns not ruled out"
```

This is not a legal standard. It is an explainable product rule that makes uncertainty visible instead of hiding it behind a green or red badge.

5.2 2. Evidence chain, not just evidence upload

Every artifact receives a provenance record: who supplied it, when it was captured, what source or consent state applies, which transformation was performed, and which version of the site layer was used. The packet can therefore answer "what did we know, from where, and when?" This is the difference between a photograph repository and a defensible safeguarding system.

5.3 3. Cultural consent and contribution trust

The cultural layer must not treat community knowledge as open raw material. Every contribution carries a visibility state such as public, community-only, reviewer-only, or withheld. A culture bearer may correct a local name, add a story, or mark a practice as seasonal without exposing sensitive details. The MVP can implement this through role-based moderation and explicit consent metadata rather than pretending that automated moderation understands cultural ownership.

5.4 4. Change ledger and temporal comparison

A single report is easy to ignore. A versioned ledger makes patterns visible. Each event records the asset, category, time, approximate location, evidence, confidence, and review status. The roadmap can add repeat photography and temporal imagery, but the MVP already demonstrates a meaningful continuity model: "what was observed, where, when, and what happened next."

5.5 5. Offline-tolerant field capture

Heritage sites may have weak connectivity. The reporting form should save a signed local draft, retain the capture timestamp and GPS metadata, and synchronize when connectivity returns. The demo can simulate this with a local queue and a visible "pending sync" state. This makes the product feel designed for field reality rather than a presentation-only dashboard.

5.6 Data contract

Entity	Minimum fields	Trust property
HeritageTwin	site_id, names, significance, practice cards, geometry, sources, consent state	The showcase object is versioned and source-labelled.
LayerVersion	geometry, source URL, source date, confidence, limitation note	Every spatial result can be reproduced.
Observation	case ID, category, neutral text, photo, coordinate, accuracy, time	Observation remains distinct from official finding.
ChangeEvent	observation link, asset link, event time, visibility, status	History accumulates without public accusation.
ReviewAction	actor role, action, note, time, next state	The workflow is auditable.

6 Architecture and implementation plan

```
Mobile PWA / multilingual field UI
        |
        v
API + authentication + consent policy
        |
        +--> Heritage Twin service: stories, practices, local names, media
        +--> Geospatial service: GeoJSON, point-in-polygon, boundary distance
        +--> Evidence service: hashes, metadata, object storage, packet export
        +--> Change ledger: events, versions, review actions, audit trail
        +--> Reviewer console: filters, map, status, request-more-information
```

A practical stack is React with Tailwind CSS for the mobile-first interface; Leaflet for map rendering; Turf.js or equivalent for geospatial operations; a relational database for site, case, consent, and audit entities; object storage for photographs and exports; and a server-side packet generator. AI is optional and deliberately subordinate. It may suggest a concern category or translate a neutral description, but it must never create an authoritative heritage fact, legal conclusion, or cultural interpretation without human review.

6.1 National scale without a fake national database

The scale strategy is a repeatable **Heritage Twin onboarding protocol**. A new cluster enters the system only after five checks: identity and cultural context, source-labelled geometry, consented content, safety and privacy review, and three spatial test points. This makes the national roadmap credible. India-wide impact comes from the repeatability of the data contract, not from claiming that an MVP has already mapped every site.

Stage	Expansion capability	Evidence of readiness
Campus pilot	One cluster, four scenarios, one reviewer workflow.	Repeatable demo and usability results.
District network	Multiple clusters using the same Heritage Twin schema.	Data quality dashboard and curator roles.
State program	Source/version registry, multilingual content, partner moderation.	Onboarding protocol and governance playbook.
National layer	Interoperable heritage intelligence across approved sources and custodians.	Authorized data agreements, not unsupported claims.

7 Impact model and measurable validation

The product's impact is not "more app users." It is improved continuity between cultural knowledge, field observation, and responsible review. The first pilot should measure both public comprehension and technical correctness.

Impact dimension	Pilot measure	Target for the prototype
Showcase quality	First-time user explains the asset and one associated living practice.	At least 4 of 5 testers understand the cluster within 60 seconds.
Spatial reliability	Known inside, outside, edge, and poor-GPS cases.	All four produce the expected explainable output.
Evidence completeness	Packet contains image, time, coordinate, accuracy, source, result, and limitation.	100% of submitted demo cases.
Trust and safety	Testers interpret "potential concern" as non-final.	No tester treats the product as a legal verdict.
Field usability	A user submits without coaching and can recover from offline state.	One complete report in under 3 minutes after onboarding.
Institutional usefulness	Reviewer can decide next action from one packet.	No manual reconstruction of core metadata.

The team should collect a small but credible evaluation set: students, one faculty member, one person familiar with the selected site, and one technically oriented tester. The key question is not whether the interface looks attractive. It is whether each person can correctly explain the distinction between observation, spatial indication, uncertainty, and official review.

8 Ethics, governance, and responsible boundaries

Heritage protection can create conflict if a product exposes private homes, names alleged offenders, or publishes culturally sensitive knowledge without consent. Heritage Guard is designed around restraint. Reports are private by default; the form discourages names and accusations; faces and unnecessary personal data are not required; sensitive practice details can be withheld; and every spatial result carries its source and limitation.

The responsibility model is explicit:

Community member: supplies an observation or cultural context.
Heritage Guard: preserves provenance, computes indicative spatial context, exposes uncertainty, and prepares a packet.
Curator or organization: reviews evidence and requests clarification.
Competent authority: makes the official administrative or legal determination.

This separation is a competitive strength. The system is useful precisely because it does not pretend that a GPS point can determine permission status, ownership, guilt, or enforcement action.

9 The 120-second national-level demo

The pitch must move from cultural meaning to technical proof, not from a login screen to a feature list.

Time	Moment	What the judges should remember
0–20 s	Living heritage opening	"This is a place and a practice, not just a pin."
20–45 s	Change capture	A visitor records a neutral observation with photo, location, time, and GPS accuracy.
45–75 s	Technical reveal	The point is tested against a source-labelled polygon; edge cases become uncertain.
75–100 s	Continuity view	The observation joins a dated change ledger with prior events and consent state.
100–120 s	Review packet	A curator receives a reproducible, non-accusatory packet with a clear next action.

9.1 Closing script

"Most heritage platforms help us discover what India has. Heritage Guard helps us understand what makes a place culturally alive and notice when its context begins to change. We build a versioned Heritage Twin with the site's story, local identity, living practice, source-labelled geometry, and current condition. A citizen can document a neutral observation. Our system preserves the evidence, checks the location against the sourced boundary, accounts for GPS uncertainty, and adds the event to a change ledger. It never declares guilt or illegality. It gives a curator or competent authority a packet they can actually review. Our innovation is the continuity layer between showcasing heritage and safeguarding it."

10 Why this is hard to ignore

The proposal is memorable because it has a precise antagonist: not "lack of awareness," but the silent loss that occurs when a heritage story, boundary, practice, or condition record is fragmented across channels. It is technically credible because the core decision can be shown live and tested with known cases. It is nationally extensible because every new cluster follows the same Heritage Twin contract. It has real-world impact because it creates an accountable path from public understanding to preservation action. And it is safe because it refuses to manufacture legal certainty.

Reviewer test	Our answer	Proof in the demo
Is this just showcasing?	No. Showcasing is the entry point to a living, versioned safeguarding record.	Heritage Twin + change ledger.
Is this just a complaint portal?	No. The system adds geometry, provenance, uncertainty, evidence packaging, and review state.	Spatial result + packet.
Is the AI making accusations?	No. AI is optional and non-authoritative; the core rule is explainable GIS.	Visible rule and disclaimers.
Can it scale?	Yes, through controlled cluster onboarding rather than an unreliable master database.	Data contract + roadmap.
What changes in the real world?	A fragmented observation becomes a traceable, review-ready preservation signal.	Case status and next action.

11 Build sequence and risk controls

The team should build in evidence order. First validate the selected cluster, its cultural context, consent state, and source-labelled geometry. Then implement the data contract and change ledger. Then build field capture, spatial reasoning, packet generation, and reviewer workflow. Only after this chain works should the team add polish, multilingual refinements, or optional AI assistance.

Risk	Control	Go/no-go signal
Weak geometry	Use one source-checked polygon; publish source date and limitations; test three points.	No spatial engine demo until validated.
Cultural overreach	Use public or consented content; allow local correction and visibility controls.	Every cultural card has a source/ consent label.
False accusation	Neutral observation language, private-by-default cases, no owner identity, no legal labels.	Safety review finds zero prohibited fields.
Demo dependency	Cache map layers, seed scenarios, and rehearse offline queue.	120-second demo works without live external services.
Feature sprawl	Protect the hero chain: context → observation → spatial reasoning → uncertainty → packet.	Every feature earns a place in that chain.

12 Final recommendation

Proceed with Heritage Guard, but change the identity from "a protection atlas with showcase context" to "a living Heritage Twin and continuity system." The showcase requirement is then met at the product's core: cultural history, local identity, living practice, and geographic context are modeled as structured, source-labelled objects. The protection capability is no longer an extra feature; it is what makes the showcase consequential.

The strongest SIH sentence is:

Heritage Guard preserves the chain between a place's cultural meaning, its physical context, and the evidence needed to safeguard both.

The team should present one site deeply rather than many sites shallowly, show the uncertainty state rather than hide it, and prove the packet and change ledger rather than list speculative integrations. That combination is niche, technically demonstrable, socially responsible, and much harder to dismiss as another heritage information app.

13 References

[1] Smart India Hackathon, official portal and overview, [https://www.sih.gov.in/].

[2] Bhuvan / NRSC and Archaeological Survey of India, "Heritage sites and Monuments" mapping application, [https://bhuvan-app1.nrsc.gov.in/culture_monuments/].

[3] Archaeological Survey of India, "Monuments/Sites," official information on centrally protected monuments, [https://asi.nic.in/pages/Monuments].

[4] UNESCO, “Involvement of communities, groups and individuals,” Intangible Cultural Heritage, [https://ich.unesco.org/en/involvement-of-communities-00033].

[5] UNESCO, “Convention for the Safeguarding of the Intangible Cultural Heritage,” Article 15 on community participation, [https://ich.unesco.org/en/convention].

[6] The supplied document, **Heritage Guard — Complete Product Blueprint for SIH 2026 PS 26197**, used as the source concept and baseline for this revision.